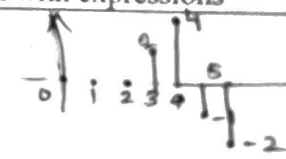




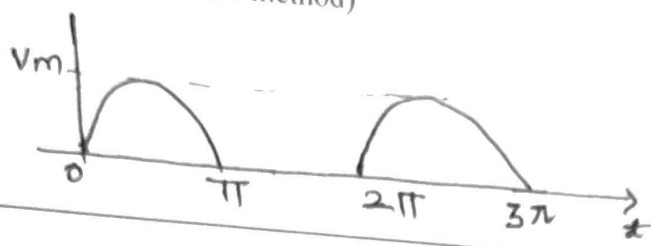
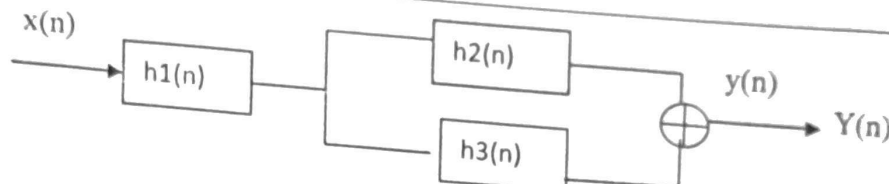
SOMAIYA
VIDYAVIHAR UNIVERSITY

12.05.2026 (E)

Semester: January 2026– April 2026		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 03	Class: SY	Semester: IV _SVU2023
Programme: BTech EXTC	Name of the department/Section/Center: EXTC	
Institute/School/Department: K. J. Somaiya School of Engineering		
Course Code: 216U03C404	Name of the Course: : Signals and systems	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	10
i)	The signal $x(t) = 2\cos(40\pi t) + \sin(60\pi t)$ is sampled at 75 Hz to get sampled signal $x[n]$. i) Check if $x(t)$ is periodic. - 2 ii) Write expression for $x[n]$ - 1 iii) What is the common period of the sampled signal $x[n]$ - 2	05
ii)	List any 5 properties of C.T. Fourier transform with expressions	05
iii)	Sketch following signal If $x[n] = 0$ if $n < 2$ $= 2n - 4$ if $2 \leq n < 4$ $= 4 - n$ if $4 \leq n \leq 6$ For above signal draw $x[-n]$ and $x[n+2]$ 	05
Q.2	Attempt any one	10
i)	A) Determine whether following signals are Energy or Power. i) $x(t) = 3 \cos 5\omega t$ - 2 ii) $X[n] = (0.5)^n u[n]$ - 3 B) Determine stability of the system whose impulse response is given by $h(t) = e^{-5t} u(t)$	5
ii)	A) All static systems are causal systems but all causal systems are Not static systems. Whether above statement is True or False? Justify with examples. B) Find C.T.F.T. for a signum function defined as $x(t) = 1$ for $t > 0$ and $x(t) = -1$ for $t < 0$	5



Q.3	Attempt any one	10
i)	Find Fourier series coefficients for a half wave rectified waveform of period 2π with maximum value of V_m. (You can use either exponential or Trigonometric series method) 	
ii)	 For the interconnected system shown above, find the overall impulse response Assume $h1[n] = [1 \ -1 \ 2 \ -3 \ 1]$, $h2[n] = [-2 \ 3 \ 2 \ 1 \ 4]$ and $h3[n] = [1 \ -2 \ 5 \ -1 \ 0 \ 2]$	
Q.4	Attempt the following	10
	A) Find Z transform and ROC for following signals a) $x[n] = [-1 \ 2 \ -3 \ 5 \ 1 \ 6]$ $\rightarrow 2$ b) $X[n] = (0.2)^n u[n] + (0.7)^n u[n]$ $\rightarrow 3$	5
	B) Find Inverse Z transform for ROC condition: $z > \frac{1}{3}$ $X(Z) = \frac{3 - (5/6)Z^{-1}}{[1 - (1/4)Z^{-1}][1 - (1/3)Z^{-1}]}$	5
Q.5	Attempt the following	10
	A) Explain the relationship between Z transform and D.T. Fourier Transform.	5
	B) Find D.T.F.T. for $x[n] = [1 \ 3 \ -3 \ 2 \ 4]$	5

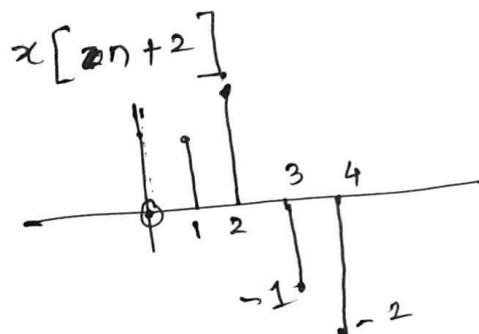
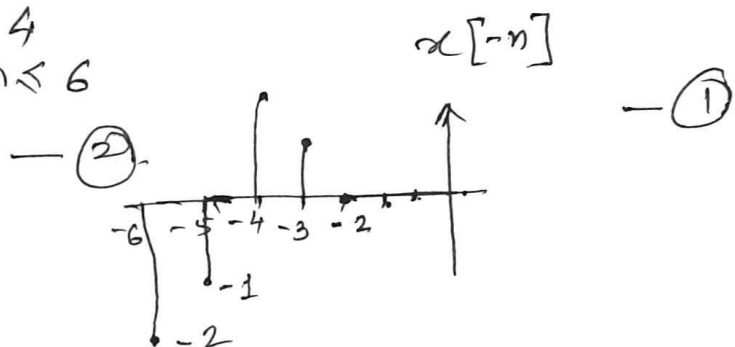
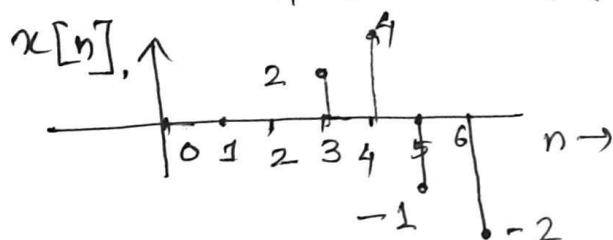
i) $x(t) = 2\cos 40\pi t + \sin 60\pi t$
 $\omega_1 = 40\pi$ $\omega_2 = 60\pi$
 $2\pi f_1 = 40\pi$ $2\pi f_2 = 60\pi$
 $\therefore f_1 = 20$ $f_2 = 30$
 $T_1 = 1/20$ $T_2 = 1/30$
 $\frac{T_1}{T_2} \rightarrow \text{Rational} \rightarrow \text{Hence periodic}$ — (2)

$x[n] = 2\cos \frac{40\pi n}{75} + \sin \frac{60\pi n}{75}$
 $= 2\cos \frac{8\pi n}{15} + \sin \frac{4\pi n}{5}$

$N_1 = 15, N_2 = 5$

Common period = LCM(5, 15) = 15

ii) $x[n] = 0 \quad n < 2$
 $= 2n - 4 \quad 2 \leq n \leq 4$
 $= 4 - n \quad 4 \leq n \leq 6$



Q2] (i) $x(t) = 3\cos 5\omega t$ → periodic signal.
 $E = \infty$ $P = \frac{1}{T} \int_0^T |x(t)|^2 dt = A^2/2 = 9/2 = 4.5$ → power signal. — (2)

$x[n] = (0.25)^n u[n]$ → Energy signal.
 $E = \sum_{n=0}^{\infty} |x(n)|^2 = \sum_{n=0}^{\infty} (0.25)^n = \frac{1}{1-0.25} = \frac{4}{3}$ — (3)

(ii) $M = \int_{-\infty}^{\infty} h(t) dt < \infty$ for stability
 $\int_0^{\infty} e^{-5t} dt = 1/5 \Rightarrow$ stable — (3)

Q3] Trigonometric series $x(t) = V_m \sin \omega t$ $0 < t < \pi$ otherwise 0

i) $a_0 = V_m / \pi$ $a_n = \frac{2V_m}{\pi(1-n^2)}$ for n even
 $= 0$ for n odd.

$b_n = 0$.

$\therefore x(t) = \frac{V_m}{\pi} - \frac{2V_m}{\pi} \left[\frac{\cos 2t}{3} + \frac{\cos 4t}{15} + \frac{\cos 6t}{35} + \dots \right]$

exponential series.

formula - (3)
 Solving - (7)

$c_0 = V_m / \pi$ $c_n = \frac{V_m}{\pi(1-n^2)}$ for n even
 $= 0$ for n odd.

ii) $h(n) = h_1(n) * [h_2(n) + h_3(n)]$

$h_2[n] = -2 \quad \downarrow \quad 3 \quad 2 \quad 1 \quad 4$

$h_3[n] = 0 \quad 1 \quad -2 \quad 5 \quad -1 \quad 0 \quad 2$

$h_2[n] + h_3[n] = [-2, 4, 0, 6, 3, 0, 2]$ $\rightarrow$ (4)

$h_1[n] * [h_2(n) + h_3(n)]$

	1	-1	2	-3	1
-2	-2	2	-4	6	-2
-4	4	-4	8	-12	4
0	0	0	0	0	0
6	6	-6	12	-18	6
3	3	-3	6	-9	3
0	0	0	0	0	0
2	2	-2	4	-6	2

$y[n] = [-2, 6, -8, 20, -17, 13, -10, 5, 7, -6, 2]$ $\rightarrow$ (6)

Q4]. $x[n] = \{ \underset{\uparrow}{-1}, 2, -3, 5, 1, 6 \}$

a) $= -1 + 2\bar{z}^{-1} - 3\bar{z}^{-2} + 5\bar{z}^{-3} + \bar{z}^{-4} + 6\bar{z}^{-5} \rightarrow \textcircled{2}$

z plane except $z=0$.

b) $(0.2)^n u(n) + (0.7)^n u(n)$

$= \frac{z}{z-0.2} + \frac{z}{z-0.7}$

$\rightarrow \textcircled{3}$

ROC $|z| > 0.2 \quad |z| > 0.7$

$\Rightarrow$ ROC $|z| > 0.7$

B] $H(z) = \frac{3 - 5/6 \bar{z}^{-1}}{(1 - 1/4 \bar{z}^{-1})(1 - 1/3 \bar{z}^{-1})} = \frac{A}{1 - \frac{1}{4} \bar{z}^{-1}} + \frac{B}{1 - \frac{1}{3} \bar{z}^{-1}}$

$\rightarrow \textcircled{3}$

$A = 1, \quad B = 2$

$X(z) = \frac{1}{1 - \frac{1}{4} \bar{z}^{-1}} + \frac{2}{1 - \frac{1}{3} \bar{z}^{-1}}$

$\therefore |z| > \frac{1}{3} \quad \left(\frac{1}{4}\right)^n u(n) + 2\left(\frac{1}{3}\right)^n u(n) \rightarrow \textcircled{2}$

Q5.

DTFT - $x(n) = \{ 1, \underset{\uparrow}{3}, -3, 2, 4 \}$

$X(z) = z + 3 - 3\bar{z}^{-1} + 2\bar{z}^{-2} + 4\bar{z}^{-3} \rightarrow \textcircled{2}$

Replace $z = e^{j\omega}$

$\therefore X(\omega) = e^{j\omega} + 3 - 3e^{-j\omega} + 2e^{-2j\omega} + 4e^{-3j\omega} \rightarrow \textcircled{3}$